MindGuard - AI Study Burnout Detector

By:

Amina Batool
62461
Irfa Arshad
63662
Nimra Atif
64813
Muqadas Khan
65314
Zarish Fatima
65315

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Human Computer Interaction

Bachelors of Science in Software Engineering

MIND GUARD

BEAT BURNOUT | STUDY SMART

Faculty of Computing
Riphah International University, Islamabad

Dedication/Acknowledgment

Thanks to Allah Almighty who made us able to complete this final project report. Also, our course teacher who guided us in this project. All the members of the team who worked hard and diligently to complete this project.

Amina Batool

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Nimra Atif

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65314

Zarish Fatima

65315

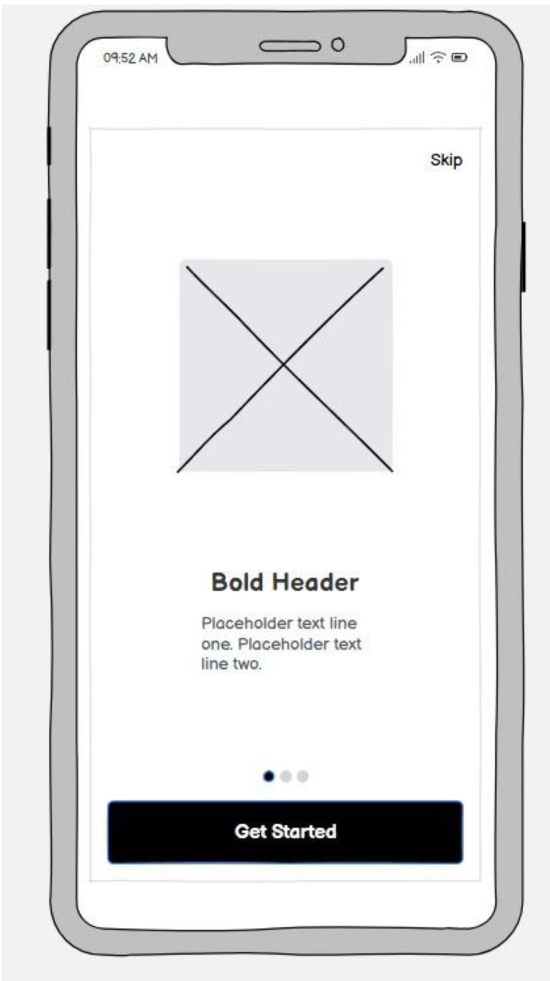
Abstract

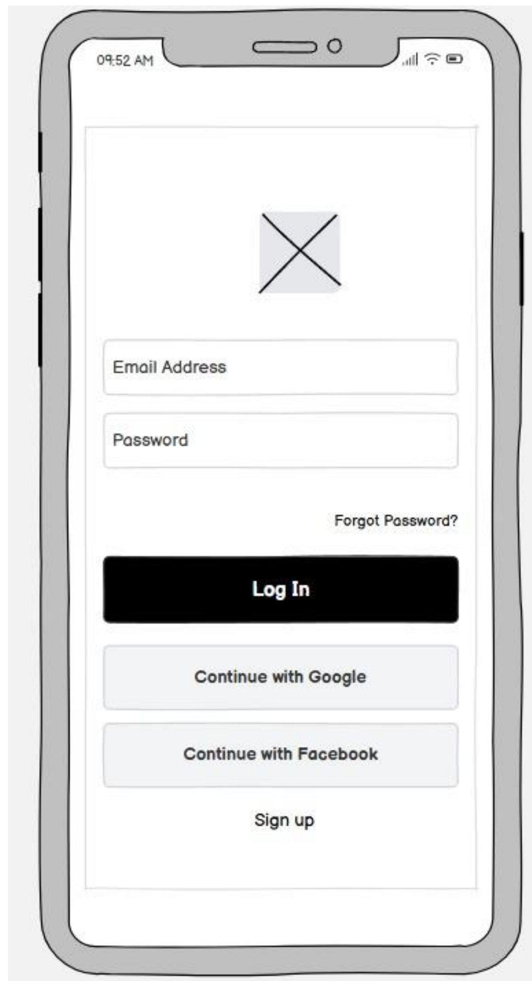
The increasing academic demands on students often result in prolonged study sessions and cognitive overload, leading to study burnout and reduced productivity. Existing productivity and wellness applications primarily focus on time management and lack mechanisms to detect mental fatigue or burnout patterns.

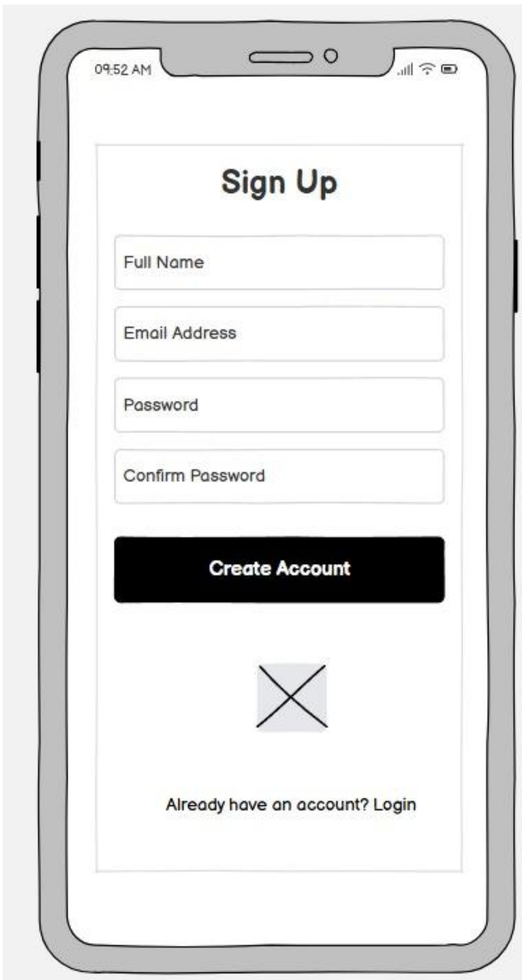
This project proposes **MindGuard**, an AI-powered study burnout detection system that analyzes user behavior, including study duration, break intervals, and self-reported mood data, to identify early signs of burnout. The system provides real-time feedback and personalized recommendations such as break scheduling, focus optimization, and wellness interventions.

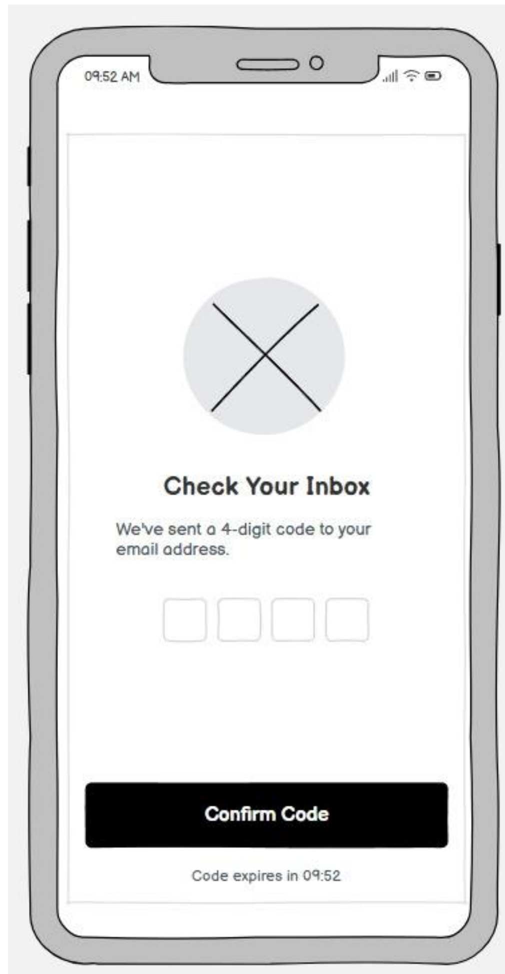
The solution is implemented as both a mobile application and a web platform, ensuring accessibility and continuous monitoring. By integrating behavioral analysis with user-centered design principles, the system aims to enhance productivity while promoting students' cognitive well-being.

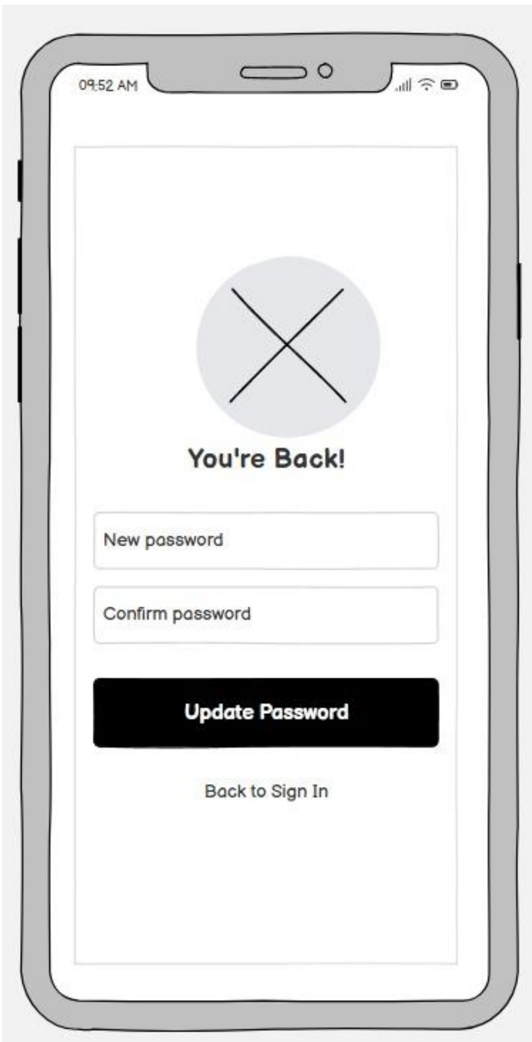
Wireframes

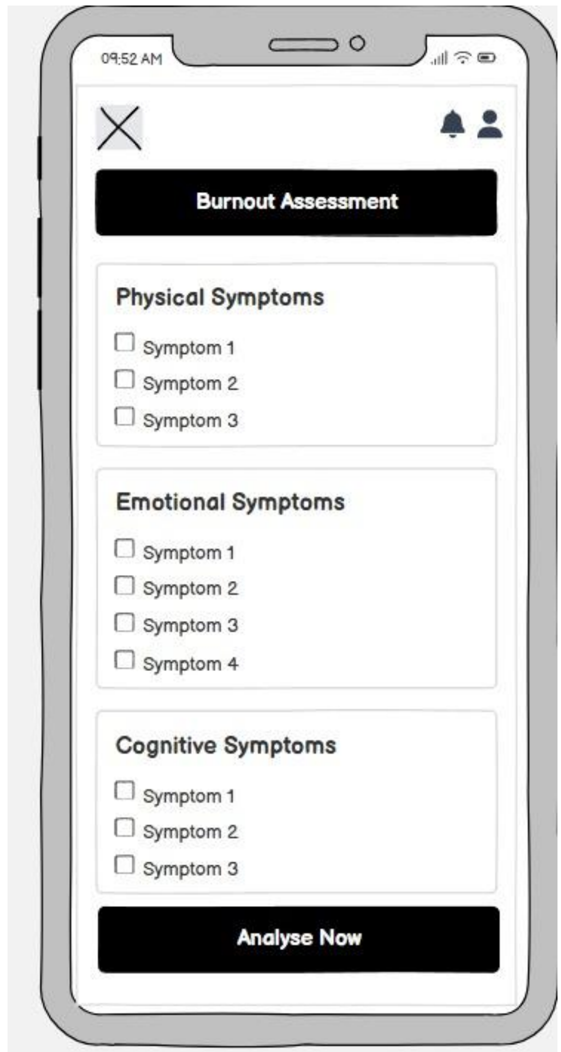

09:52 AM
Skip
Bold Header
Placeholder text line one. Placeholder text line two.
Get Started

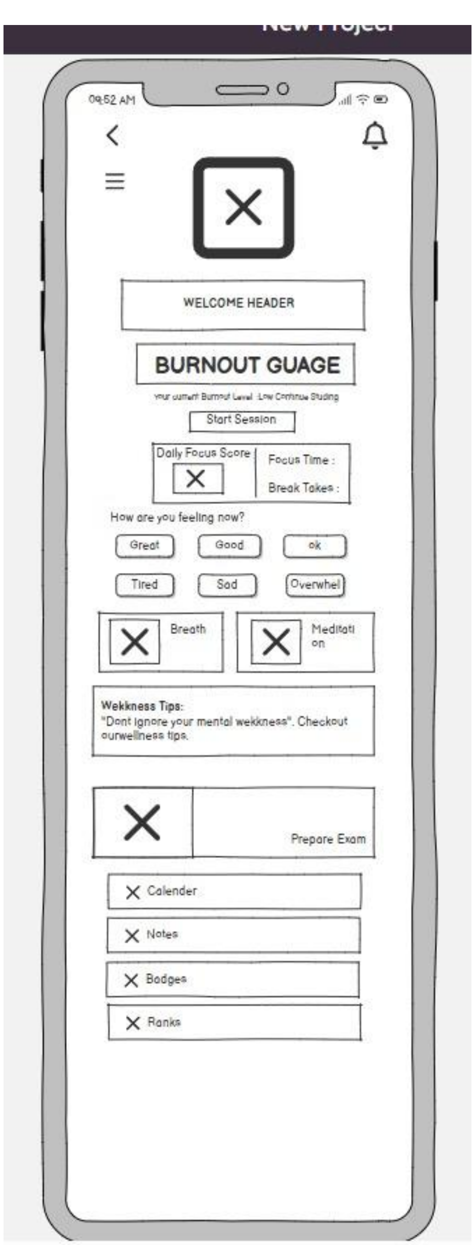

09:52 AM
Email Address
Password
Forgot Password?
Log In
Continue with Google
Continue with Facebook
Sign up

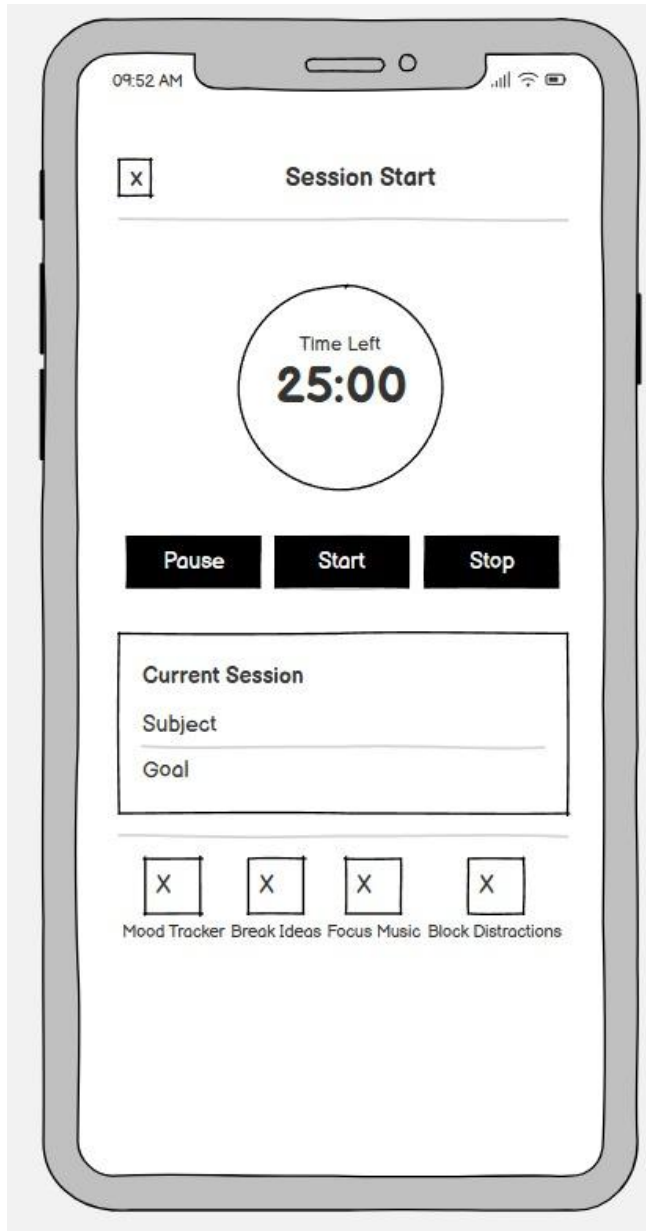

09:52 AM
Sign Up
Full Name
Email Address
Password
Confirm Password
Create Account
Already have an account? Login

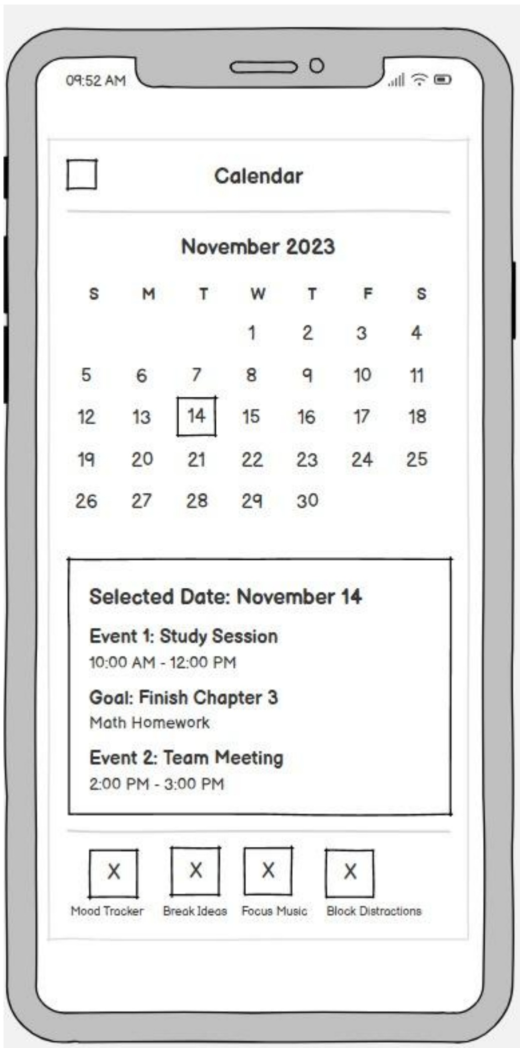

09:52 AM
Check Your Inbox
We've sent a 4-digit code to your email address.
Confirm Code
Code expires in 09:52

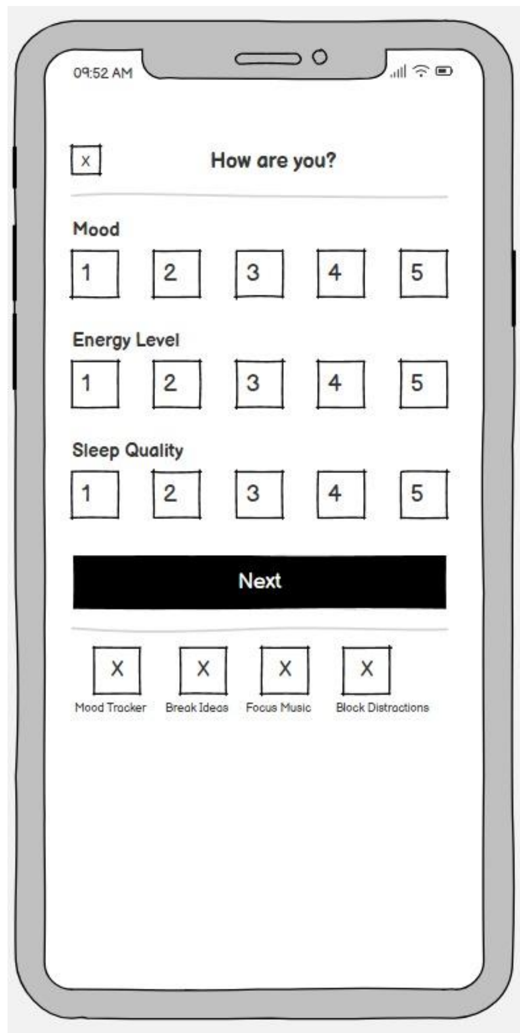

09:52 AM
You're Back!
New password
Confirm password
Update Password
Back to Sign In

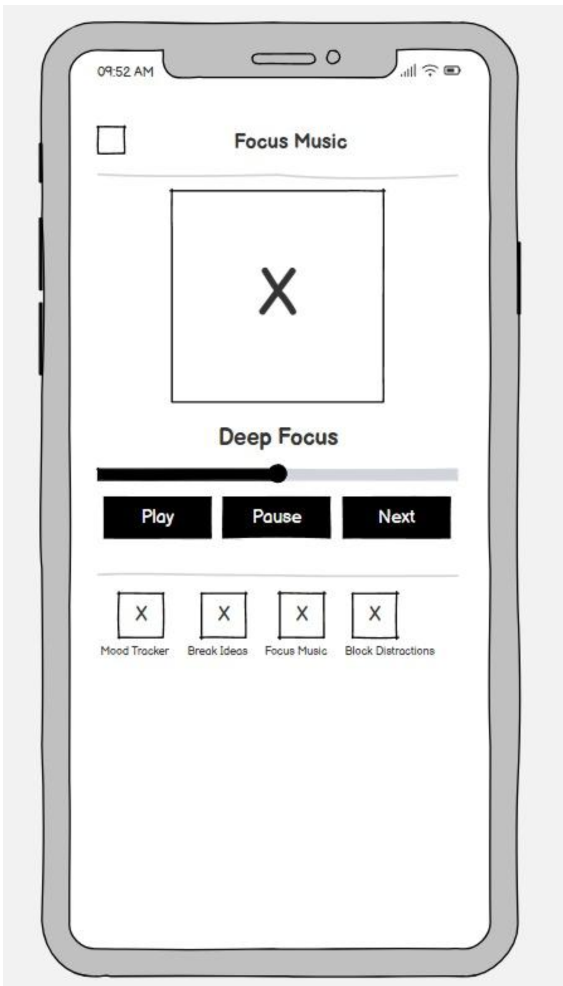
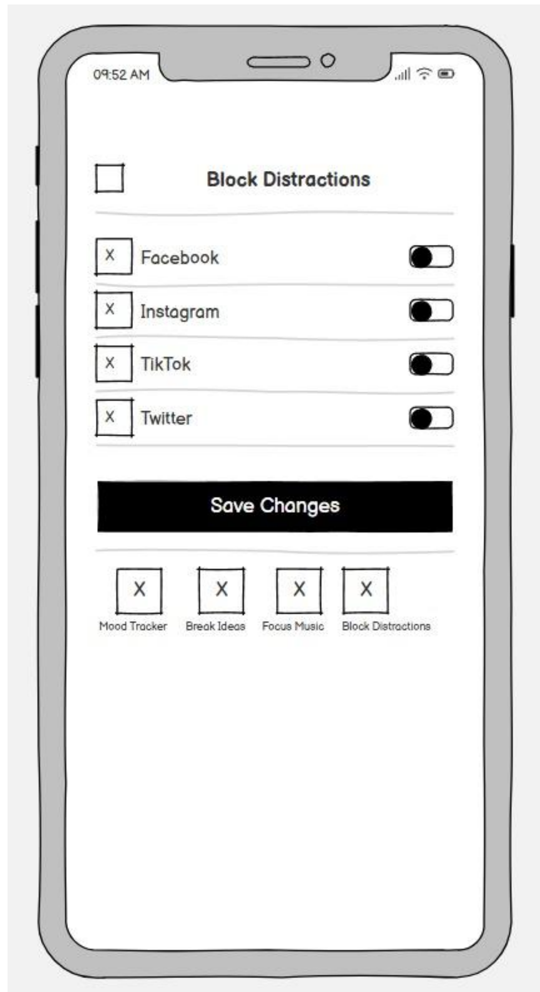

09:52 AM
Burnout Assessment
Physical Symptoms
Symptom 1
Symptom 2
Symptom 3
Emotional Symptoms
Symptom 1
Symptom 2
Symptom 3
Symptom 4
Cognitive Symptoms
Symptom 1
Symptom 2
Symptom 3
Analyse Now

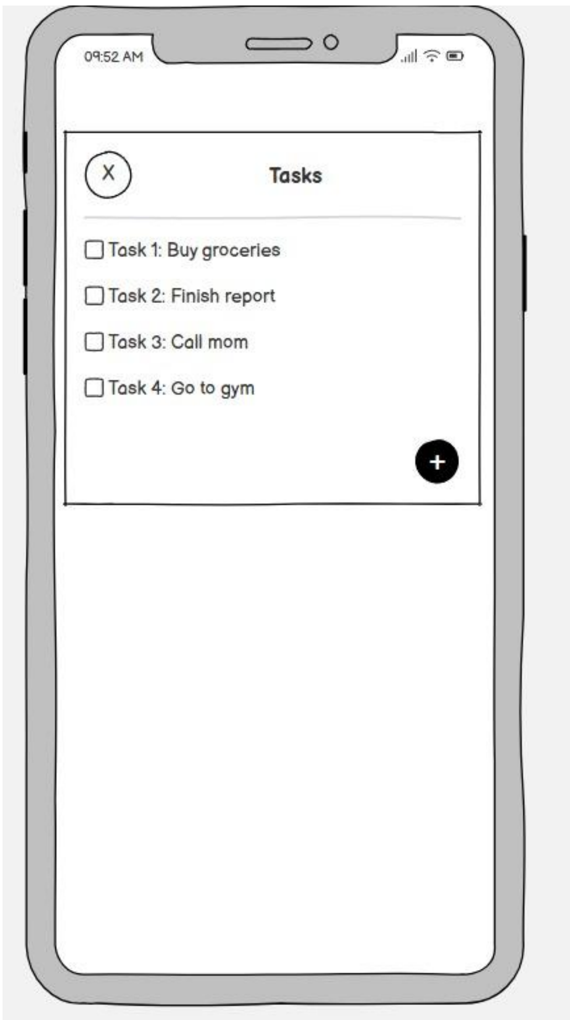

09:52 AM
WELCOME HEADER
BURNOUT GUAGE
Start Session
Daily Focus Score
Focus Time :
Break Takes :
How are you feeling now?
Great
Good
ok
Tired
Sad
Overwhel
Breath
Meditati on
Wekkness Tips:
"Dont ignore your mental wekkness". Checkout ourwellness tips.
Prepare Exam
Calender
Notes
Badges
Ranks

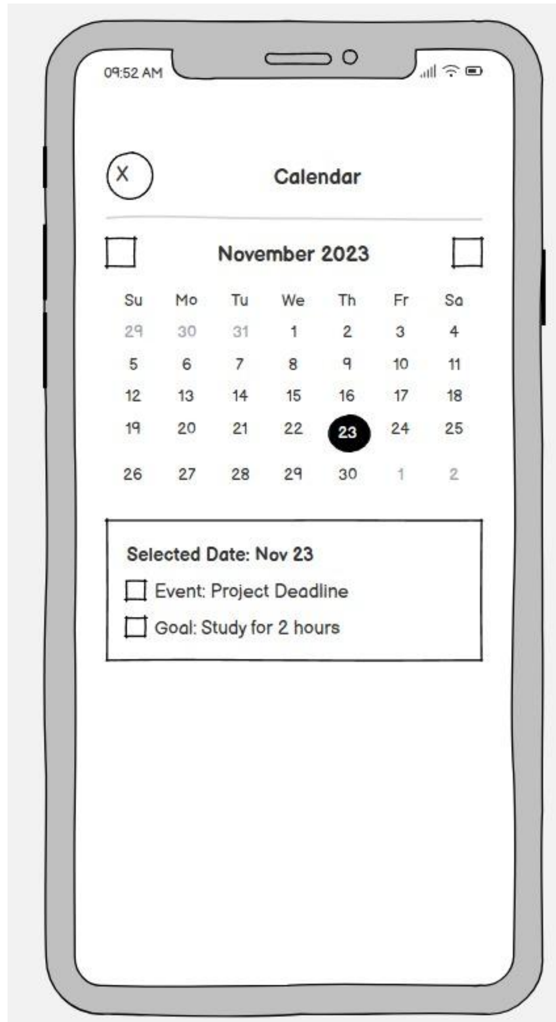

09:52 AM
Session Start
Time Left
25:00
Pause
Start
Stop
Current Session
Subject
Goal
Mood Tracker
Break Ideas
Focus Music
Block Distractions

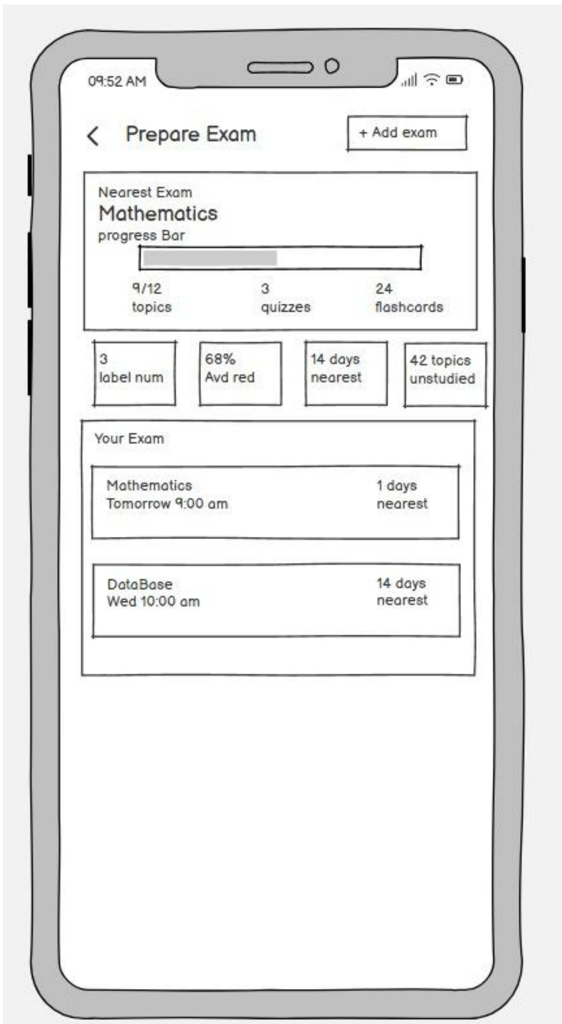

09:52 AM
Calendar
November 2023
S M T W T F S
1 2 3 4
5 6 7 8 9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30
Selected Date: November 14
Event 1: Study Session
10:00 AM - 12:00 PM
Goal: Finish Chapter 3
Math Homework
Event 2: Team Meeting
2:00 PM - 3:00 PM
X Mood Tracker
X Break Ideas
X Focus Music
X Block Distractions

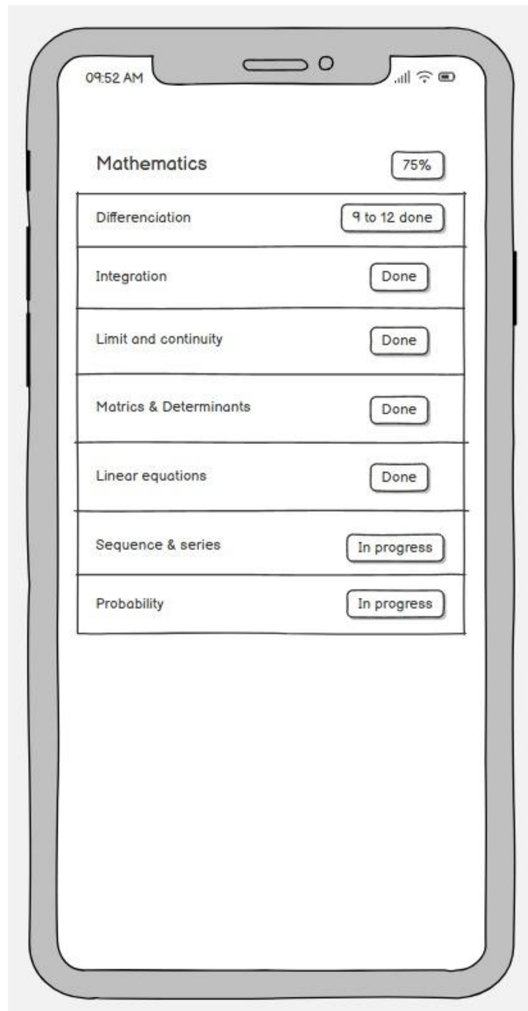

09:52 AM
X
How are you?
Mood
1 2 3 4 5
Energy Level
1 2 3 4 5
Sleep Quality
1 2 3 4 5
Next
X Mood Tracker
X Break Ideas
X Focus Music
X Block Distractions

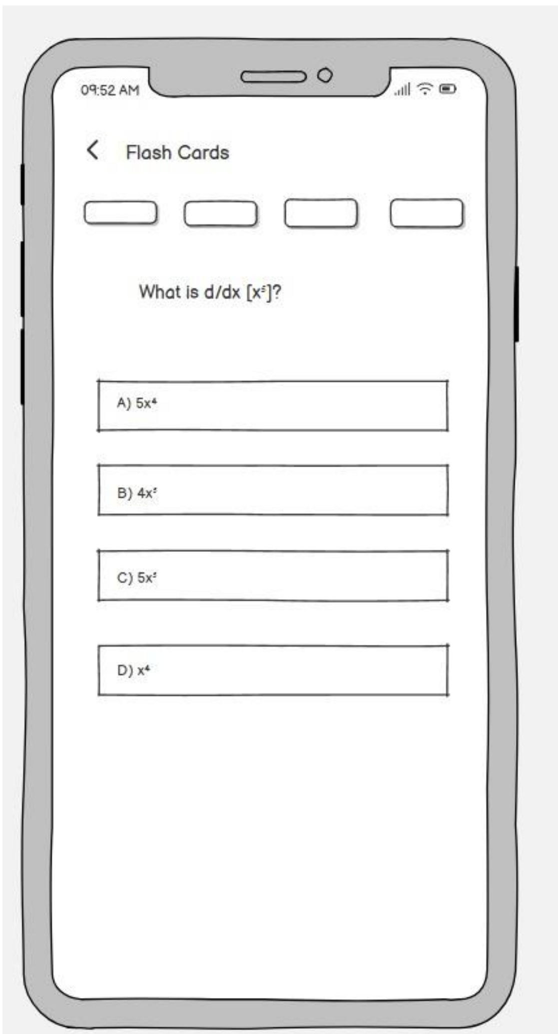

09:52 AM
Focus Music
Deep Focus
Play
Pause
Next
Mood Tracker
Break Ideas
Focus Music
Block Distractions

09:52 AM
Block Distractions
Facebook
Instagram
TikTok
Twitter
Save Changes
Mood Tracker
Break Ideas
Focus Music
Block Distractions


09:52 AM
X
Tasks
Task 1: Buy groceries
Task 2: Finish report
Task 3: Call mom
Task 4: Go to gym
+

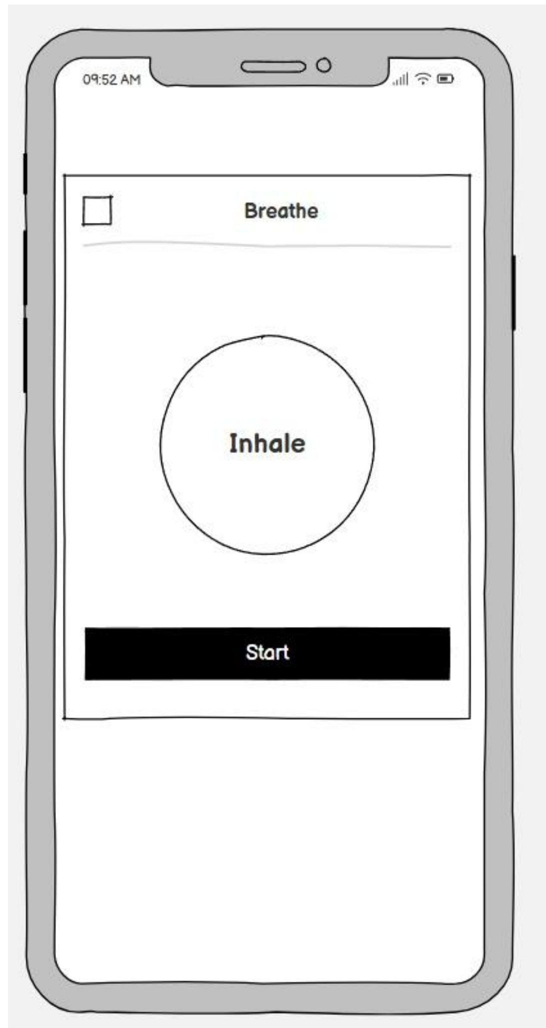

09:52 AM
X
Calendar
November 2023
Su Mo Tu We Th Fr Sa
29 30 31 1 2 3 4
5 6 7 8 9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 1 2
Selected Date: Nov 23
Event: Project Deadline
Goal: Study for 2 hours


09:52 AM
Prepare Exam
+ Add exam
Nearest Exam
Mathematics
progress Bar
9/12
topics
3
quizzes
24
flashcards
3
label num
68%
Avd red
14 days
nearest
42 topics
unstudied
Your Exam
Mathematics
Tomorrow 9:00 am
1 days
nearest
DataBase
Wed 10:00 am
14 days
nearest

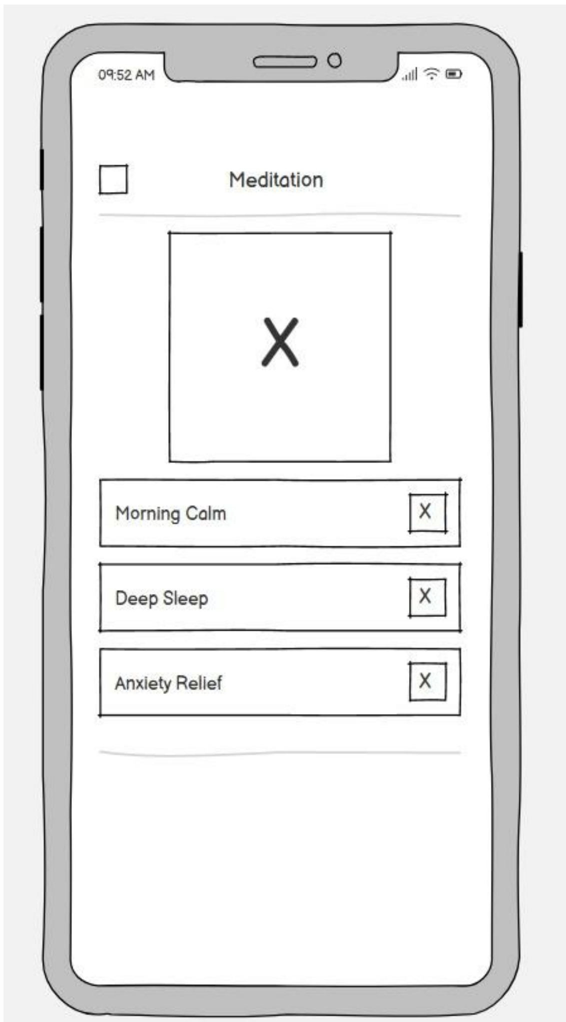

09:52 AM
Mathematics
75%
Differenciation
9 to 12 done
Integration
Done
Limit and continuity
Done
Matrics & Determinants
Done
Linear equations
Done
Sequence & series
In progress
Probability
In progress

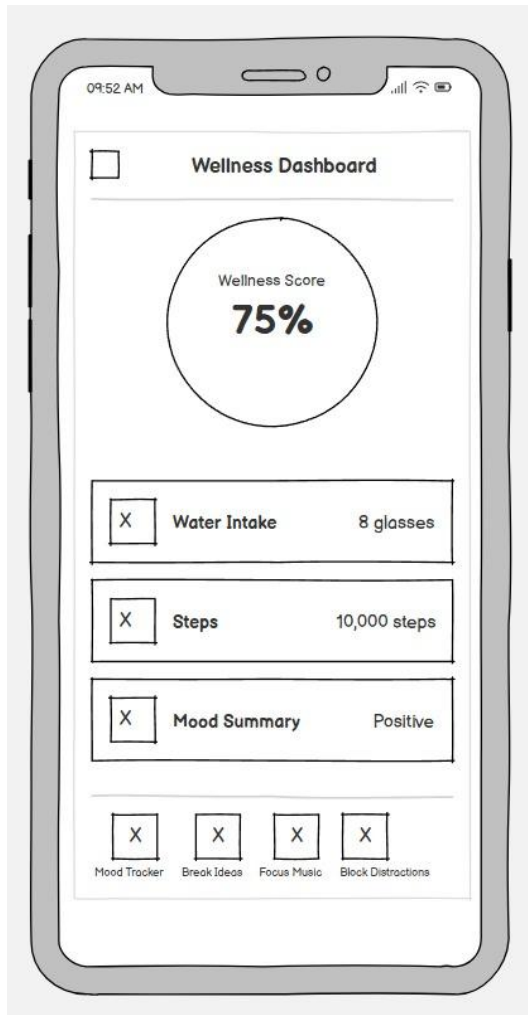


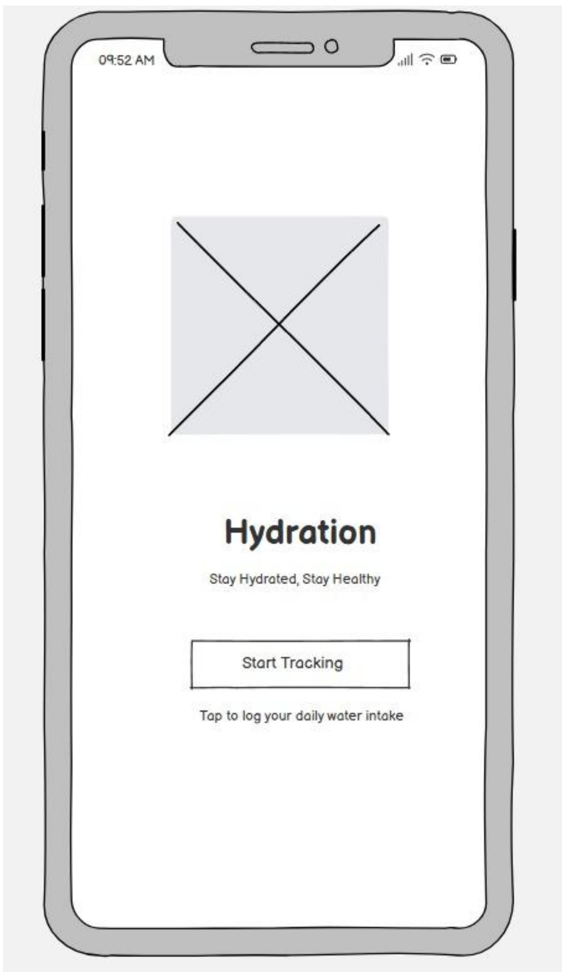
09:52 AM
Flash Cards
What is d/dx [x⁵]?
A) 5x⁴
B) 4x⁵
C) 5x⁵
D) x⁴

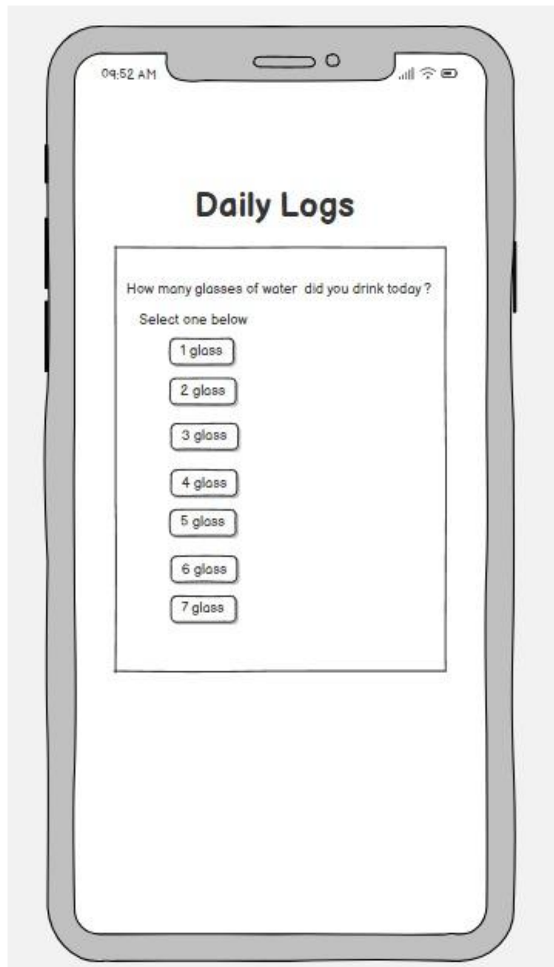


09:52 AM
Breathe
Inhale
Start


09:52 AM
Meditation
X
Morning Calm
X
Deep Sleep
X
Anxiety Relief
X

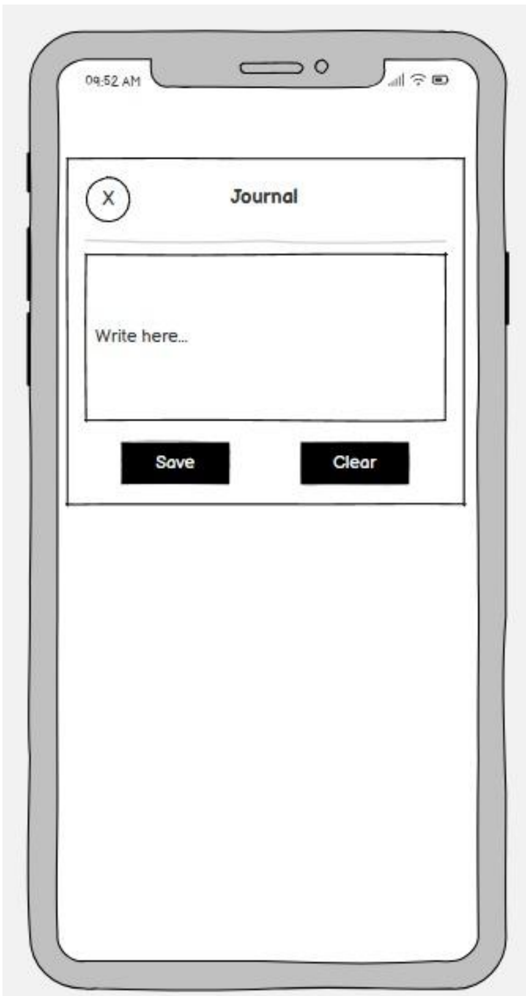

09:52 AM
Wellness Dashboard
Wellness Score
75%
X
Water Intake
8 glasses
X
Steps
10,000 steps
X
Mood Summary
Positive
X
X
X
X
Mood Tracker
Break Ideas
Focus Music
Block Distractions

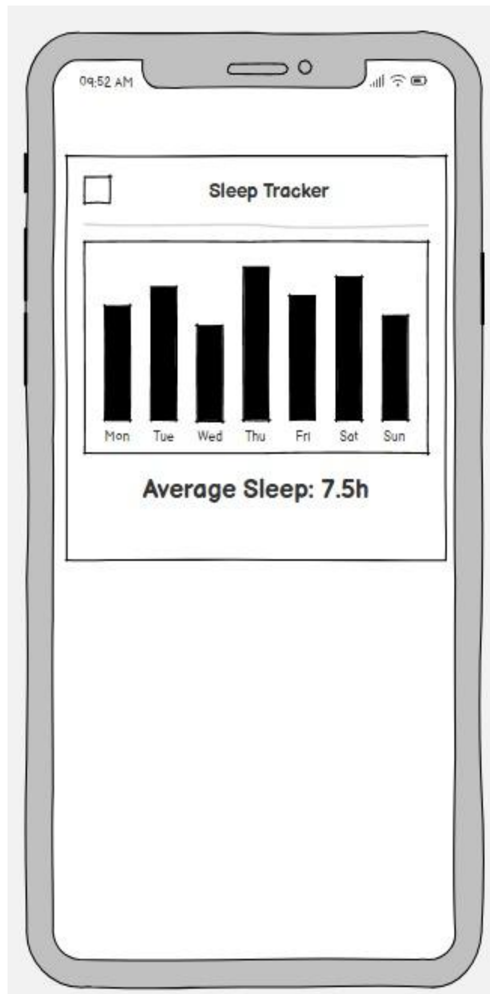

09:52 AM
Hydration
Stay Hydrated, Stay Healthy
Start Tracking
Tap to log your daily water intake


09:52 AM
Daily Logs
How many glasses of water did you drink today ?
Select one below
1 glass
2 glass
3 glass
4 glass
5 glass
6 glass
7 glass

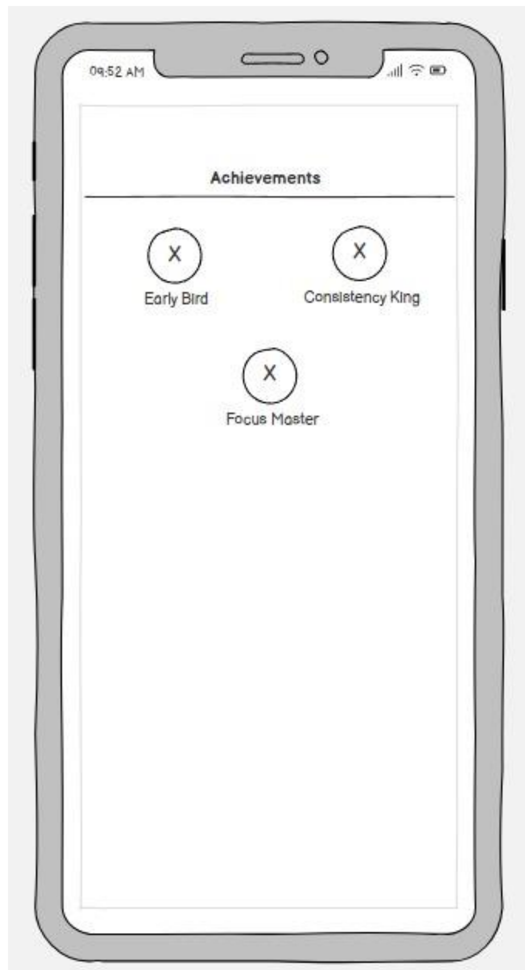

09:52 AM
Notes
Search...
Note Title 1
This is a snippet of the first note content...
Note Title 2
This is a snippet of the second note content...
Note Title 3
This is a snippet of the third note content...
Note Title 4
This is a snippet of the fourth note content...
Note Title 5
This is a snippet of the fifth note content...
X
Tasks
X
Calendar
X
Notes
X
Journal


09:52 AM
X
Notes
Search...
Meeting Notes
Discussed Q4 strategy and budget allocations...
Idea for new app
A productivity app with gamification features...
Shopping List
Milk, eggs, bread, coffee, vegetables...

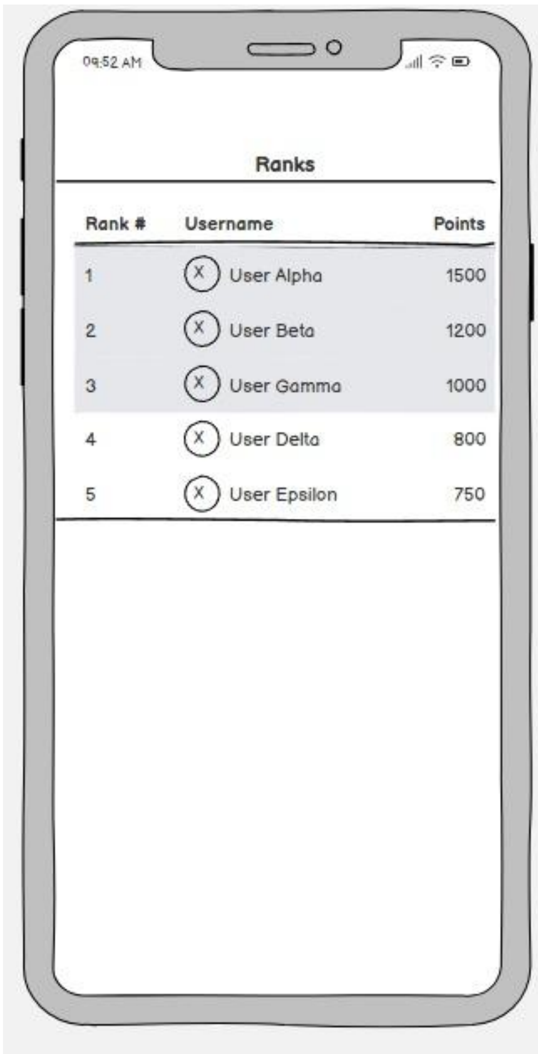

09:52 AM
X
Journal
Write here...
Save
Clear

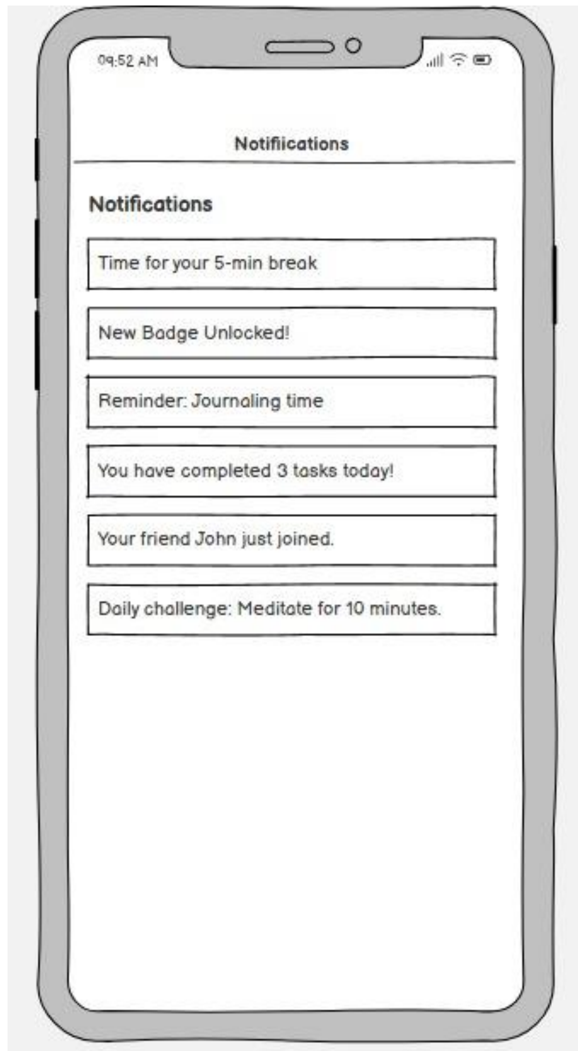

09:52 AM
Sleep Tracker
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Average Sleep: 7.5h

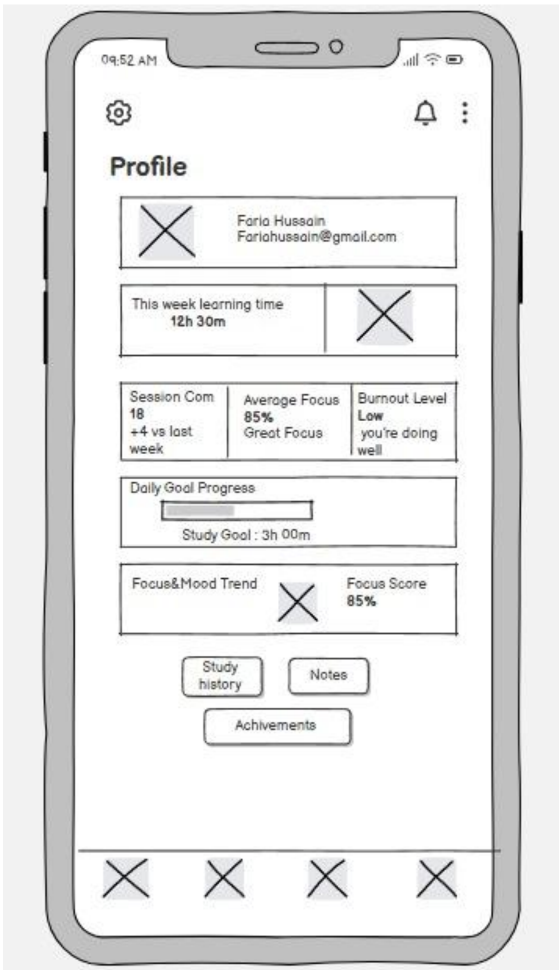
09:52 AM
5-Minute Break
05:00
Motivational Quote Placeholder
Skip



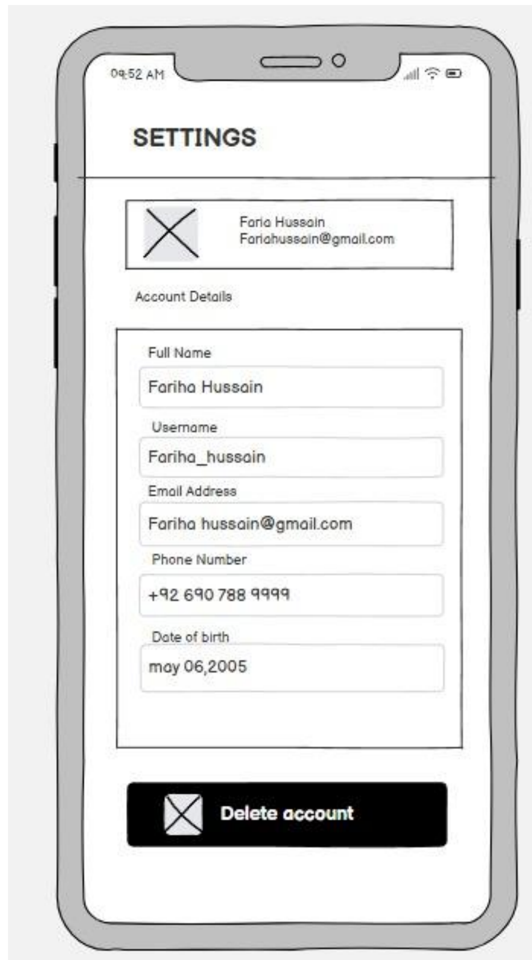
09:52 AM
Achievements
X
Early Bird
X
Consistency King
X
Focus Master


09:52 AM
Ranks
Rank #
Username
Points
1
User Alpha
1500
2
User Beta
1200
3
User Gamma
1000
4
User Delta
800
5
User Epsilon
750

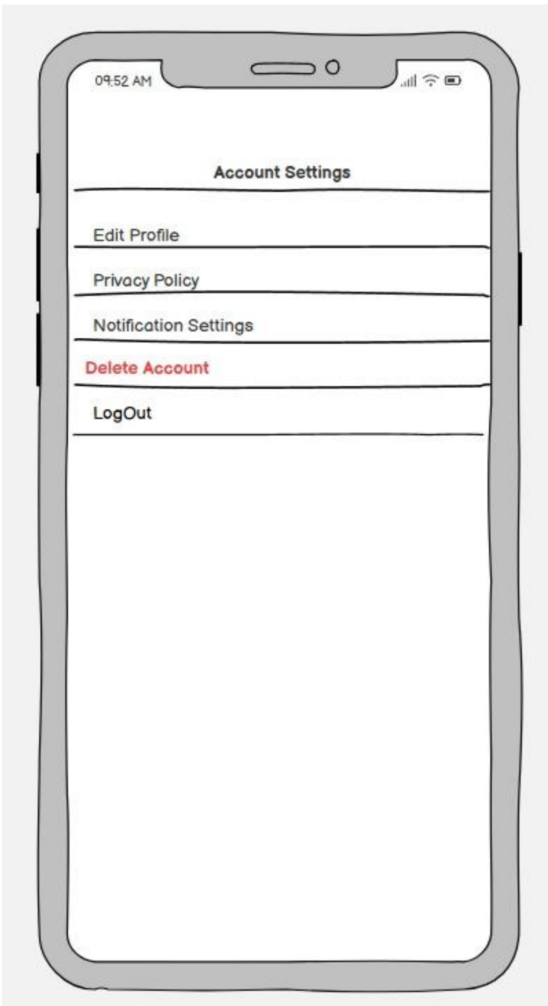

09:52 AM
Notifiications
Notifications
Time for your 5-min break
New Badge Unlocked!
Reminder: Journaling time
You have completed 3 tasks today!
Your friend John just joined.
Daily challenge: Meditate for 10 minutes.

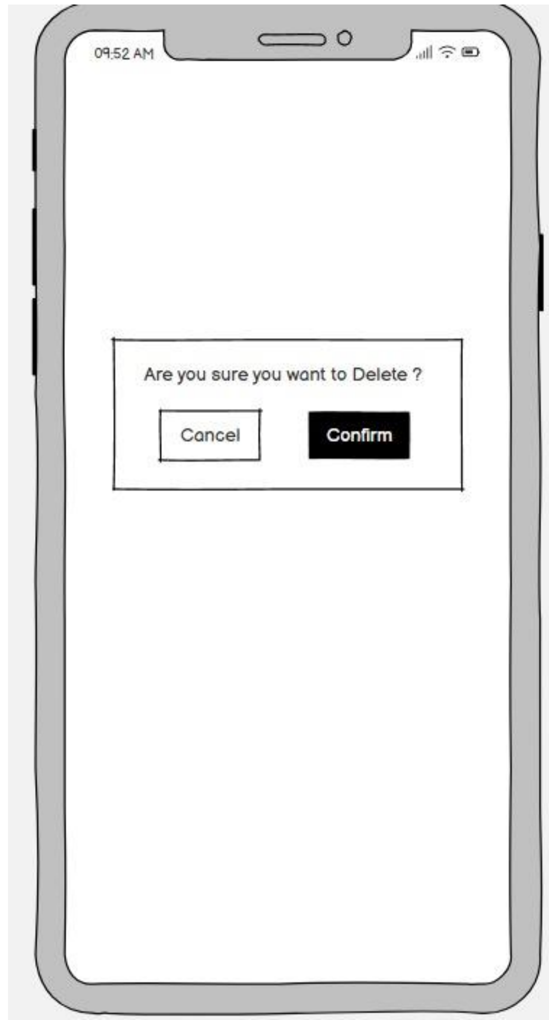


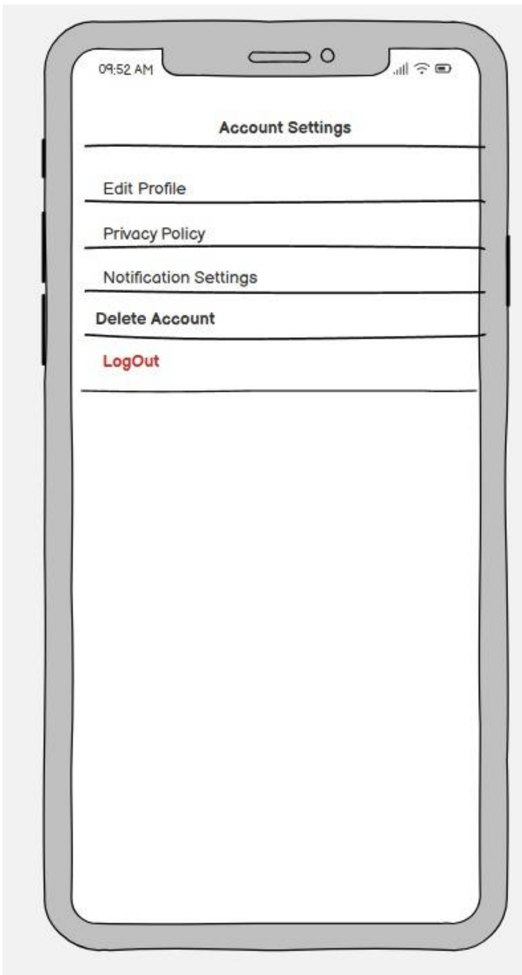
04:52 AM
Profile
Faria Hussain
Fariahussain@gmail.com
This week learning time
12h 30m
Session Com
18
+4 vs last
week
Average Focus
85%
Great Focus
Burnout Level
Low
you're doing
well
Daily Goal Progress
Study Goal : 3h 00m
Focus&Mood Trend
Focus Score
85%
Study
history
Notes
Achivements



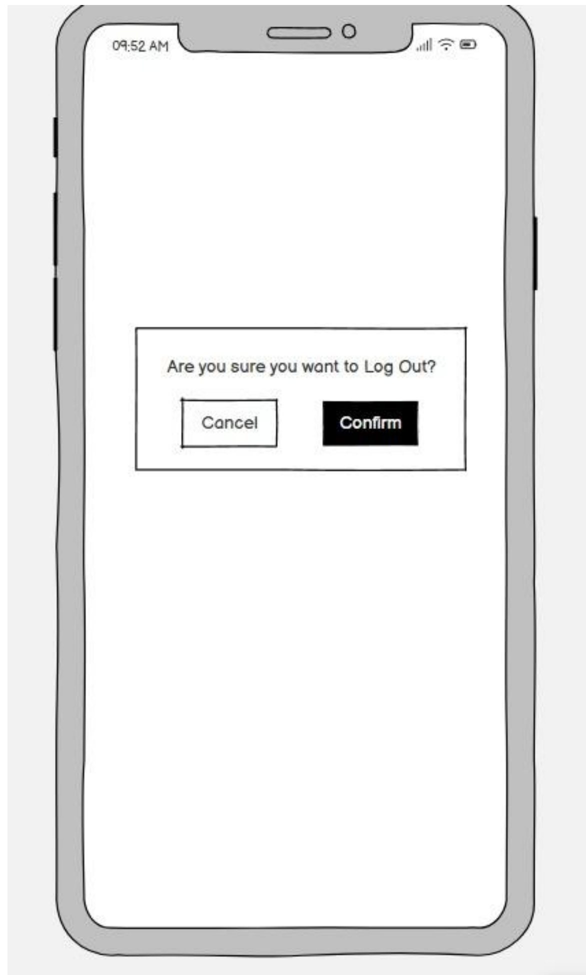
04:52 AM
SETTINGS
Faria Hussain
Fariahussain@gmail.com
Account Details
Full Name
Fariha Hussain
Username
Fariha_hussain
Email Address
Fariha hussain@gmail.com
Phone Number
+92 690 788 9999
Date of birth
may 06,2005
Delete account


09:52 AM
Account Settings
Edit Profile
Privacy Policy
Notification Settings
Delete Account
LogOut


09:52 AM
Are you sure you want to Delete ?
Cancel
Confirm



09:52 AM
Account Settings
Edit Profile
Privacy Policy
Notification Settings
Delete Account
LogOut



09:52 AM
Are you sure you want to Log Out?
Cancel
Confirm